

Blue Jay Night Ride — Proposal Speaker Notes

Team FourSight • EN.553.602 • Target ~12–14 min total + Q&A. Each presenter ~3 min. Say the words in **Say**; the Cue line is what to point at. End on your **handoff** line.

Irfan — Slides 1–6 Intro & the service (~3 min)

Slide 1 — Cover

Say: "Good evening — we're Team FourSight, and our project is optimizing the Blue Jay Night Ride,

Hopkins' late-night on-demand shuttle." Keep it to one line and advance.

Image-only cover — adjust the line to whatever title/graphic is on it.

Slide 2 — Team & course

Say: "Quick introductions. We're four students across Applied Math and Data Science — Miles, Irfan,

Youzhi, and Vignesh — working in Professor Kearsley's Research & Design course. Each of us owns a

piece of the pipeline, from the data to the optimization model."

Slide 3 — Visual lead-in

Say: "This is the service we're focused on: the Blue Jay Night Ride shuttle that runs around the

Homewood campus after dark."

Image-only slide — tailor to the actual photo/graphic shown.

Slide 4 — Project title: Optimizing the Blue Jay Night Ride

Say: "Our goal in one sentence: dispatch and routing models that cut rider wait times and reduce the trips that spill over to Lyft. Everything today builds toward that."

Slide 5 — How the shuttle works

Say: "How it works, in three steps: a rider books in the TransLoc app; a shared van is assigned and may pool other riders on the way; and you're dropped curb-to-curb. It runs roughly 6 PM to 2 AM in the Homewood area."

Cue: walk the Request → Pool → Curb-to-curb steps.

Slide 6 — The service's promises (our constraints)

Say: "These published promises become the hard constraints in our model: pickup within 20 minutes — 25 when busy — rides under 25 minutes, groups up to five, and the van waits three minutes at pickup. It's safety-first, so distance alone doesn't decide the answer. And when no van can keep the promise, the trip is handed to Lyft as paid overflow — hold onto that, it comes back in a minute."

Handoff: "Miles will show what the data says about how well those promises are being kept."

Miles — Slides 7-10 Motivation & problem framing (~3 min)

Slide 7 — Motivation: the demand

Say: "Here's the scale. In April, 44,374 riders used the service. Demand is heavy and sharply peaked

— about 5,100 rides in the 6 PM hour, with roughly 28 vans out — and it's spread across the city. The

key takeaway: supply already scales to the peak, so this is a smarter-assignment problem, not a just-

add-vans problem."

Cue: point to the 6 PM peak on the bar chart and the origin-density map.

Slide 8 — Motivation: the gap (the key slide)

Say: "Most riders are fine — the typical wait and ride are both about 11 minutes. The problem is the

tail. The slowest one in ten — the 90th percentile — wait 17 to 21 minutes and ride 21 to 30, right at or

past the targets. Read the bars left to right: green is under target, amber is close, red is over. And on

the right, only about 73% of trips get a van — roughly one in four overflow to a paid Lyft. That tail, plus

the overflow, is exactly what smarter dispatch can shrink."

Cue: define P90 out loud using the explainer box; point to the red bands and the donut.

Slide 9 — Why it's hard: Dial-a-Ride, not ordinary routing

Say: "Why is this hard? It's not standard routing. Each request is a paired pickup and drop-off that the

same van must serve, pickup before drop-off. The van's load changes as people get on and off, and

requests arrive live while some moves are already committed. That combination is the dynamic Dial-a-

Ride problem — a step beyond a normal vehicle-routing problem."

Cue: contrast the CVRP diagram (left) with the paired-pickup/drop-off Dial-a-Ride diagram (right).

Slide 10 — Data sources

Say: "On data: today we have aggregate TransLoc and JHU dashboards — origin-destination density,

rides by hour and status, vans in service, and wait- and ride-time percentiles. We've requested de-

identified trip-level extracts — the timestamps, zones, party size, and vehicle IDs. Aggregates let us

calibrate demand and validate; trip-level data lets us replay real nights. Everything stays de-identified

— no names, IDs, or exact addresses."

Handoff: "Youzhi will take us into the methods."

Youzhi — Slides 11-14 Methods & demo (~4 min)

Slide 11 — Technical strategy: a spectrum of methods

Say: "We compare a spectrum of methods on the same request stream. Fast heuristics — nearest-

feasible and greedy insertion — are our speed baseline and reuse our prior routing work. The

centerpiece is an exact rolling-horizon mixed-integer program that dispatches under all the constraints.

And a hybrid — the MILP plus local search — is our scaling path for the busy 28-van peak. Same inputs

everywhere, so any difference is the policy, not luck."

Slide 12 — The model at the core

Say: "At the core, the model decides the route and who rides together. The yes/no choices are:

which van, which road segment, or send it to Lyft. The continuous quantities are arrival times, van load,

wait, and extra ride time. The objective is service-first — first cut Lyft overflow, then rider wait, then

excess ride time, and only then van travel. Service before shaving miles."

Cue: point to the priority list; keep the math light — full notation is backup.

Slide 13 — Making it dynamic (rolling horizon)

Say: "The model solves a single moment, but real dispatch keeps changing, so we wrap it in a loop:

observe the current state, freeze what's already underway, re-solve the rest under a short time limit,

and commit the next move — then re-optimize when the next request arrives. That's how a static

model becomes a live dispatcher without ever yanking a pickup that's already in progress."

Cue: trace the four steps — Observe → Freeze → Resolve → Commit.

Slide 14 — Proof of concept (the demo)

Say: "To show the pipeline works, here's a simulation on real streets, routed with OSRM. Left is

greedy dispatch — longer detours, and a request overflows to Lyft. Right is optimized — tighter

pooling, shorter waits, no overflow, on the exact same requests. This is illustrative with synthetic

demand, not final results, but it proves we can build and visualize the whole loop."

Cue: if presenting live, switch to the browser demo (open in Chrome/Safari) and press play.

Handoff: "Vignesh will cover how we'll judge it and the plan."

Vignesh — Slides 15–17 Evaluation, plan & close (~3 min)

Slide 15 — How we'll judge it

Say: "We evaluate fairly: every method sees identical inputs — the same request stream, fleet, travel

times, and starting state. We compare reconstructed-historical, nearest-feasible, greedy insertion, the

rolling-horizon MILP, and the hybrid — on metrics that match the problem: wait median and P90,

percent of pickups within 20 minutes, ride time and percent within 25, percent overflow to Lyft, miles

and utilization, and solver time. It's honest by design — if a simple rule matches the MILP, that's a valid

finding. And if trip-level data is delayed, we fall back to calibrated synthetic demand with multiple

seeds."

Slide 16 — Plan & scope

Say: "Here's the semester plan: data audit, travel-time matrix, simulator, baselines, the MILP, rolling-horizon control, experiments, and the report. We've de-risked it — if data is late we use calibrated synthetic demand; if the MILP is too slow we prune, warm-start, or go hybrid; if we can't reconstruct history we compare transparent heuristics. And scope: this is a retrospective decision-support study of what causes long waits and overflow and whether optimization can reduce them. It will not touch live vans or TransLoc operations."

Cue: run along the milestone timeline; keep it brisk.

Slide 17 — Close / Q&A

Say: "To sum up: most riders are fine, but the slowest tenth and the heavy Lyft overflow are real, and we believe smarter dispatch can move both — safely, as decision support. Thank you; we're happy to take questions."

Image-only closing — adjust to whatever's on the final slide (thank-you / Q&A / team).